

DATA MANIPULATION WITH PYTHON

Fundamentals, Numpy, Pandas

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Outline

- 1 Foundations of Python
- 2 Numerical Analysis with Numpy
- 3 Data Manipulation with Pandas



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Basic Definitions in Python

- **Python** is a **high-level**, **interpreted**, and **general-purpose** programming language.
- **High-level** means that **Python** is designed to be easy to read and write.
- **Interpreted** means that **Python** code is executed line by line, rather than being compiled into machine code.
- **Weakly typed** means that **Python** does not require you to declare the type of a variable.
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Popular Developer Tools

Life is Short, Use Dev Tools

ByteByteGo



Development Env



Diagramming



AI Tools



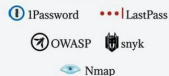
Hosting & Deployment



Code Quality



Security



Note-taking



Design



Virtual Environments

- **Virtual environments** are a way to create isolated spaces on your computer for **Python projects**.
- **Virtual environments** allow you to **install packages** and **dependencies** for a specific project without affecting other projects.
- **Virtual environments** are created using the **venv** module, which is included in the Python standard library.
- **Virtual environments** are activated using the **source** command in the terminal.
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Modules and Packages

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- **Modules** are used to **organize code** and **make it reusable**.
- **Packages** are **directories** that contain **Python files (modules)**.
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Dependencies Management

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- **Dependencies** are traditionally managed using a **pyproject.toml** file, which lists the **names** and **versions** of the packages required by your project.
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Variables and Memory Management

Variables Definition

Variables are used to store data values. Python has no command for declaring a variable. A variable is created the moment you first assign a value to it.



Conditionals

Definition

Conditionals are used to execute different code blocks based on *different conditions*.

Nested Conditionals

Nested conditionals are conditionals that are *inside* other conditionals.

Elif Conditionals

Elif conditionals are used to check *multiple conditions*.



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Loops and Range

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The **range function** is used to generate a sequence of numbers.



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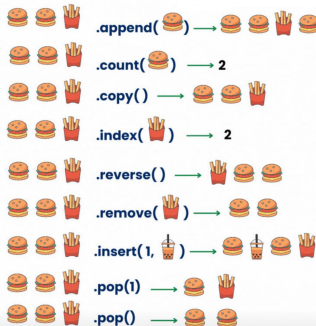


Lists

Definition

A **list** is a collection of items that are *ordered* and *changeable*. **Lists** are defined by enclosing the items in square brackets.

PYTHON LIST METHODS



Dictionaries

Definition

A **dictionary** is a collection of items that are *unordered*, *changeable*, and *indexed*. Dictionaries are defined by enclosing the items in curly braces.



Sets and Tuples

Definition Sets

A **set** is a collection of items that are *unordered* and *unindexed*. **Sets** are defined by enclosing the items in curly braces.

Definition Tuples

A **tuple** is a collection of items that are *ordered* and *unchangeable*. **Tuples** are defined by enclosing the items in parentheses.



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Lists Comprehensions

Definition

List comprehensions provide a concise way to create lists. Common applications are to make *new lists* where each element is the result of some operation applied to each member of another sequence or iterable, or to create a subsequence of those elements that satisfy a certain condition.



Functions

Definition

A **function** is a block of code that only runs when it is called. You can pass data, known as parameters, into a function. A **function** can return data as a result.

Type of Functions

- Built-in Functions
- User-defined Functions
 - Variadic Functions
 - Recursive Functions



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Python Iterators

Definition

An **iterator** is an object that contains a *countable number of values*. An **iterator** is an object that can be iterated upon, meaning that you can traverse through all the values.

Maps

The **map function** is used to apply a function to *all the items* in an input list.

Filters

The **filter function** is used to *select items* from an input list that meet a certain condition.



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Lambda Functions

Definition

A **lambda function** is a small anonymous function. A **lambda function** can take any number of arguments, but can only have one expression.



Classes and Objects

Definition

Python is an *object-oriented programming* language. Almost everything in Python is an **object**, with its *properties* and *methods*. A **class** is like an object constructor, or a “blueprint” for creating objects.



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Numerical Python Library — Numpy

- **Numpy** is the **core** library for **scientific computing** in Python. It is the **fundamental package** for scientific computing with **Python**.
- **Numpy** is a general-purpose **array-processing** package. It provides a **high-performance** multidimensional array object, and tools for working with these arrays.
- **Numpy** was created by **Travis Oliphant** in 2005, and it is an **open-source project**. Currently, *Numpy version 2.5.1* is available.



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Lineal Algebra with Numpy

- **Numpy** provides a *comprehensive set* of **linear algebra** functions.
- **Numpy** provides the *functionality* to *create* and **manipulate matrices**.
- **Numpy** provides the *functionality* to **solve linear systems** of equations.
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Vectorization with Numpy

- **Vectorization** is the *process* of **converting** an **algorithm** from **operating** on a single value at a time to **operating** on a set of values at one time.
- **Vectorization** is the *process* of **replacing** explicit **loops** with **array expressions** or **matrix operations**.
- The *advantages* of **vectorization** are **speed** and **clarity**. The *disadvantages* are **memory** and **complexity**.
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Typical Operations with Numpy

- **Numpy** provides the *functionality* to **create** and **manipulate arrays**.
- **Numpy** provides the *functionality* to **perform element-wise operations** on **arrays**, **matrix operations** on **arrays**.
- **Numpy** provides the *functionality* to **perform linear algebra operations** on **arrays**.
- **Numpy** provides the *functionality* to **perform statistical operations** on **arrays**.



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Introduction to Pandas

- **Pandas** is a *fast, powerful, flexible, and easy-to-use* **open-source data manipulation** and **data analysis** library built on top of the Python programming language.
- **Pandas** is a **high-level data manipulation** tool developed by *Wes McKinney* in 2008.
- **Pandas** is a **fast and efficient data manipulation** tool that is *built* on top of NumPy.
- **Pandas** is one of the most **popular and widely-used data manipulation** libraries in the *world*.



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The “Series” Data Structure

- A **Series** is a **one-dimensional array-like object** that contains a *sequence of values* and an associated array of data labels, called the **index**.
- The **index** of a **Series** is an *array of labels* that correspond to the **values** in the **Series**. The **index** of a **Series** is an *optional parameter* that *defaults* to a sequence of integers starting at **zero**.
- The **Series object** is a **core data structure** in **Pandas**.



Querying a Series

- You could **query** a **Series** using **indexing** (boolean or fancy).
- You could **query** a **Series** using `loc` and `iloc` indexers.



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The “DataFrame” Data Structure

- A **DataFrame** is a **two-dimensional labeled data structure** with *columns* of potentially different types.
- A **DataFrame** is a **tabular data structure** that is *similar* to a spreadsheet or a SQL table.
- A **DataFrame** is a **core data structure** in **Pandas**. It is a two-dimensional size-mutable data structure with **labeled axes** (rows and columns).
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DataFrame Indexing and Loading

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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/data-analysis-programming>

